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# PersonGen: Knowledge-Preserving Personalization of Text-to-Image Diffusion Models

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## Abstract

Recent text-to-image diffusion models demonstrate remarkable generative capabilities but require adaptation to generate personalized concepts. Fine-tuning approaches such as DreamBooth and LoRA allow efficient personalization, but may alter previously learned knowledge and reduce generalization ability.

In this work, we study the problem of knowledge-preserving personalization of text-to-image diffusion models. We combine LoRA-based personalization with AlphaEdit-inspired model editing methods and investigate how personalized concept information is represented inside diffusion models.

We analyze the contribution of cross-attention key/value parameters to concept transfer and study whether editing these components is sufficient for preserving the original model capabilities while introducing new visual concepts.

**Keywords:** text-to-image diffusion models, personalization, LoRA, DreamBooth, model editing, Stable Diffusion, knowledge preservation

## 1 Introduction

Text-to-image diffusion models have recently achieved impressive results in high-quality image generation. However, adapting these models to new concepts, objects, or subjects remains a challenging problem. A personalized model should learn the target concept while preserving the knowledge acquired during pretraining.

Existing personalization methods, such as DreamBooth and LoRA, efficiently adapt pretrained diffusion models using a small number of images. However, fine-tuning can introduce unintended modifications and reduce the ability of the model to generate previously learned concepts.

An alternative direction is model editing, where model parameters are modified in a constrained manner to introduce new knowledge while minimizing changes to the original model. This work investigates whether such approaches can be combined with personalization methods for more controlled concept transfer.

The main goal of this research is to analyze how personalized concepts are stored in text-to-image diffusion models and whether knowledge-preserving editing methods can improve personalization quality.

**Contributions.** Our contributions can be summarized as follows:

- We develop a pipeline combining DreamBooth LoRA personalization with AlphaEdit-based model editing for Stable Diffusion models.
- We analyze the role of cross-attention key/value matrices in storing personalized concepts.

- We perform ablation studies over textual identifiers, LoRA training parameters, checkpoints, and editing configurations.
- We investigate the trade-off between concept strength and preservation of pretrained model knowledge.

**Outline.** The rest of the paper is organized as follows. Section 2 describes related work. Section 3 introduces preliminary concepts. Section 4 presents the proposed approach. Section 5 describes experimental evaluation. Section 6 discusses the obtained results.

## 2 Related Work

### Text-to-image personalization.

Personalization methods adapt pretrained diffusion models to generate specific subjects or concepts using a limited number of examples. DreamBooth introduces subject-driven generation by fine-tuning model parameters, while LoRA provides a parameter-efficient alternative by learning low-rank updates.

### Knowledge-preserving model editing.

Model editing methods aim to modify neural networks while minimizing unwanted changes to existing knowledge. AlphaEdit introduces constrained editing strategies that project parameter updates into subspaces preserving previous model behavior.

### Concept representation in diffusion models.

Previous studies analyze how textual representations and attention mechanisms affect image generation. In this work, we focus on cross-attention key/value matrices and their role in transferring personalized concepts.

## 3 Preliminaries

### 3.1 General notation

Let  $M_0$  denote a pretrained Stable Diffusion model and  $M_p$  denote a personalized model obtained after fine-tuning.

The LoRA personalization process can be represented as:

$$W_{LoRA} = W_0 + \Delta W,$$

where  $W_0$  is the original parameter matrix and  $\Delta W$  is a low-rank adaptation.

### 3.2 Assumptions

We assume that:

- the pretrained diffusion model contains general visual knowledge;
- personalization modifies only a subset of model parameters;
- cross-attention layers play an important role in connecting textual representations with visual concepts.

## 4 Method

The proposed pipeline consists of several stages.

### 4.1 LoRA personalization

First, DreamBooth LoRA fine-tuning is performed to learn a new visual concept from a small image dataset. A special identifier token is introduced to represent the personalized concept.

## 4.2 Attention parameter extraction

After personalization, cross-attention key and value projection matrices are extracted from the diffusion model.

These matrices are analyzed as possible carriers of personalized information.

## 4.3 Text representation analysis

We construct text embedding matrices corresponding to:

- identifier tokens;
- content words;
- all valid tokens.

## 4.4 AlphaEdit-based modification

The extracted representations are used to compute constrained parameter updates. The goal is to transfer the personalized concept while minimizing changes to the original model.

# 5 Experiments

We conduct experiments using Stable Diffusion v1-4.

The experimental evaluation includes:

- comparison of different identifier tokens;
- LoRA checkpoint analysis;
- learning rate ablations;
- comparison of different text representation strategies;
- interpolation between LoRA and edited weights.

For interpolation experiments we define:

$$W(\alpha) = (1 - \alpha)W_{edit} + \alpha W_{LoRA},$$

where  $\alpha = 0$  corresponds to the edited model and  $\alpha = 1$  corresponds to the original LoRA personalization.

Evaluation is performed using generated image comparisons and analysis of concept preservation.

# 6 Discussion

The experiments demonstrate that personalized information is distributed across multiple components of diffusion models.

Although modifying cross-attention parameters can transfer part of the concept, the full personalization behavior cannot always be recovered using only key/value matrices.

This suggests that successful personalization requires considering a broader set of parameters responsible for concept representation.

# 7 Conclusion

In this work, we investigate knowledge-preserving personalization of text-to-image diffusion models.

We propose a research pipeline combining DreamBooth LoRA personalization and AlphaEdit-based editing. Our analysis provides insights into the localization of personalized knowledge inside diffusion models and the challenges of performing controlled concept transfer.

## References

## **A Appendix / supplemental material**

### **A.1 Additional experiments / Proofs of Theorems**

Additional experimental results, generated image grids, and ablation studies are provided in the supplementary material.